

Random Variable

A **random variable** is a variable whose value depends on the outcome of a random experiment. Instead of assigning names like Head or Tail, it assigns **numerical values** to the outcomes.

Definition

A **random variable (X)** is a function that maps each outcome in the sample space to a real number.

In simple words:

A random variable converts the outcomes of a random experiment into numbers so that we can perform mathematical calculations like finding probabilities, mean, and variance.

Example 1: Tossing a Coin

Experiment: Toss one coin

Sample Space (S):

$$S = \{ H, T \}$$

Define a random variable X:

- $X = 1$ if Head occurs
- $X = 0$ if Tail occurs

Outcome	Random Variable (X)
H	1
T	0

Here, **X** is the random variable.

Example 2: Rolling a Die

Experiment: Roll a fair die.

Sample Space:

$S = \{ 1, 2, 3, 4, 5, 6 \}$

Let X = Number shown on the die

Outcome	X
1	1
2	2
3	3

4	4
5	5
6	6

Here the random variable simply represents the face value.

Example 3: Number of Heads in Two Coin Tosses

Experiment: Toss two coins.

Sample Space:

$S = \{ HH, HT, TH, TT \}$

Define

$X = \text{Number of Heads}$

Outcome	Number of Heads (X)
HH	2

HT	1
TH	1
TT	0

So the possible values of X are:

$$X = \{ 0, 1, 2 \}$$

Types of Random Variables

1. Discrete Random Variable

A random variable that takes **countable** values.

Examples:

- Number of heads in 5 coin tosses
- Number of students in a classroom
- Number of defective products

Possible values:

0,1,2,3,...

Characteristics

- Countable outcomes
- Probability Mass Function (PMF)
- Probability of a specific value can be greater than zero

2. Continuous Random Variable

A random variable that can take **any value within an interval**.

Examples:

- Height of a person
- Weight
- Temperature
- Time taken to finish an exam

Possible values:

Any value between two numbers, such as

150.1, 150.25, 150.256, 150.256

Characteristics

- Infinite possible values
 - Uses Probability Density Function (PDF)
 - Probability at one exact value is zero
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Why Do We Need Random Variables?

Random variables help us:

- Convert random outcomes into numerical values.
 - Calculate probabilities easily.
 - Find statistical measures such as **mean**, **variance**, and **standard deviation**.
 - Build probability distributions.
 - Apply machine learning and statistical models.
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Quick Comparison

Discrete Random Variable	Continuous Random Variable
Countable values	Infinite values
PMF	PDF
Number of students	Height of students
Dice outcome	Temperature
Number of calls	Time taken to answer a call

Key Points to Remember

- A random variable assigns **numbers** to outcomes of a random experiment.
- It is denoted by capital letters such as **X, Y, Z**.
- There are **two types**:

- **Discrete Random Variable** (countable values)
- **Continuous Random Variable** (measurable values)
- Random variables are the foundation for probability distributions and many statistical techniques.

1. Joint Probability

Definition

Joint probability is the probability that **two events occur at the same time**.

Notation

$P(A \cap B)$

Read as:

"Probability of A **and** B"

Example

Roll a die.

Let:

- A = Getting an even number = {2,4,6}
- B = Getting a number greater than 3 = {4,5,6}

Common outcomes = {4,6}

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

Keyword: AND

2. Marginal Probability

Definition

Marginal probability is the probability of **a single event occurring**, regardless of the other event.

It is obtained by summing (marginalizing) the joint probabilities over the other variable.

Notation

$P(A)$

or

$P(B)$

Example

Roll a die.

Probability of getting an even number:

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

Here we ignore every other event.

Keyword: Single event

3. Conditional Probability

Definition

Conditional probability is the probability of one event occurring **given that another event has already occurred**.

Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example 1: Rolling a Die ★ (Easiest)

A fair die is rolled.

Let:

- A = Getting an even number = {2, 4, 6}
- B = Getting a number greater than 3 = {4, 5, 6}

Since we already know the number is greater than 3, the possible outcomes become:

{4,5,6}

Among these, the even numbers are:

{4,6}

Therefore,

- $P(B) = \frac{3}{6}$

- $P(A \cap B) = \frac{2}{6}$

$$P(A|B) = \frac{\frac{2}{6}}{\frac{3}{6}}$$

Answer: $\frac{2}{3}$

Example 2: Drawing a Card

A card is drawn from a standard deck of 52 cards.

Find the probability that the card is a **King**, given that it is a **Face Card**.

Face cards:

- Jack = 4
- Queen = 4
- King = 4

Total face cards = 12

Kings = 4

$$P(\text{King} | \text{Face Card}) = \frac{4}{12}$$

Answer: $\frac{1}{3}$

Example 3: Rain and Traffic

Suppose:

- 40 days had rain.
- Out of those 40 rainy days, traffic was heavy on 30 days.

Find the probability of heavy traffic given that it rained.

$$P(\text{Traffic} | \text{Rain}) = \frac{30}{40}$$

Answer: 75%

Example 4: Medical Testing

A hospital tested 200 patients.

- 50 have diabetes.
- Among those 50 patients, 35 have high blood pressure.

Find:

Probability that a patient has high blood pressure **given that** the patient has diabetes.

$$P(\text{BP} | \text{Diabetes}) = \frac{35}{50}$$

Answer: 70%

Example 5: Machine Learning (Spam Detection)

A dataset contains 1000 emails.

- Spam emails = 200
- Emails containing the word "Lottery" = 80
- Spam emails containing "Lottery" = 60

Find:

Probability that an email is spam given that it contains the word "Lottery."

$$P(\text{Spam} | \text{Lottery}) = \frac{60}{80}$$

Answer: 75%

Three unbiased coins are tossed simultaneously. What is the conditional probability that at least two coins show heads, given that at least one coin shows heads?



Two fair six-sided dice are rolled. What is the conditional probability that the sum of the numbers rolled is 7, given that the first die shows an even number?

Die 2	6	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)
	5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
	4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
	3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
	2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
	1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
		1	2	3	4	5	6
		Die 1					

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Comparison Table

Probability Type	Meaning	Symbol	Keyword
Joint Probability	Both events occur together	$P(A \cap B)$	AND
Marginal Probability	Probability of one event	$P(A)$	Single event
Conditional Probability	Probability of A given B	$P(A B)$	GIVEN